

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN CLIMATE

MODULE - 3

INDIAN MONSOON

- **FACTORS DETERMINING THE CLIMATE OF INDIA**

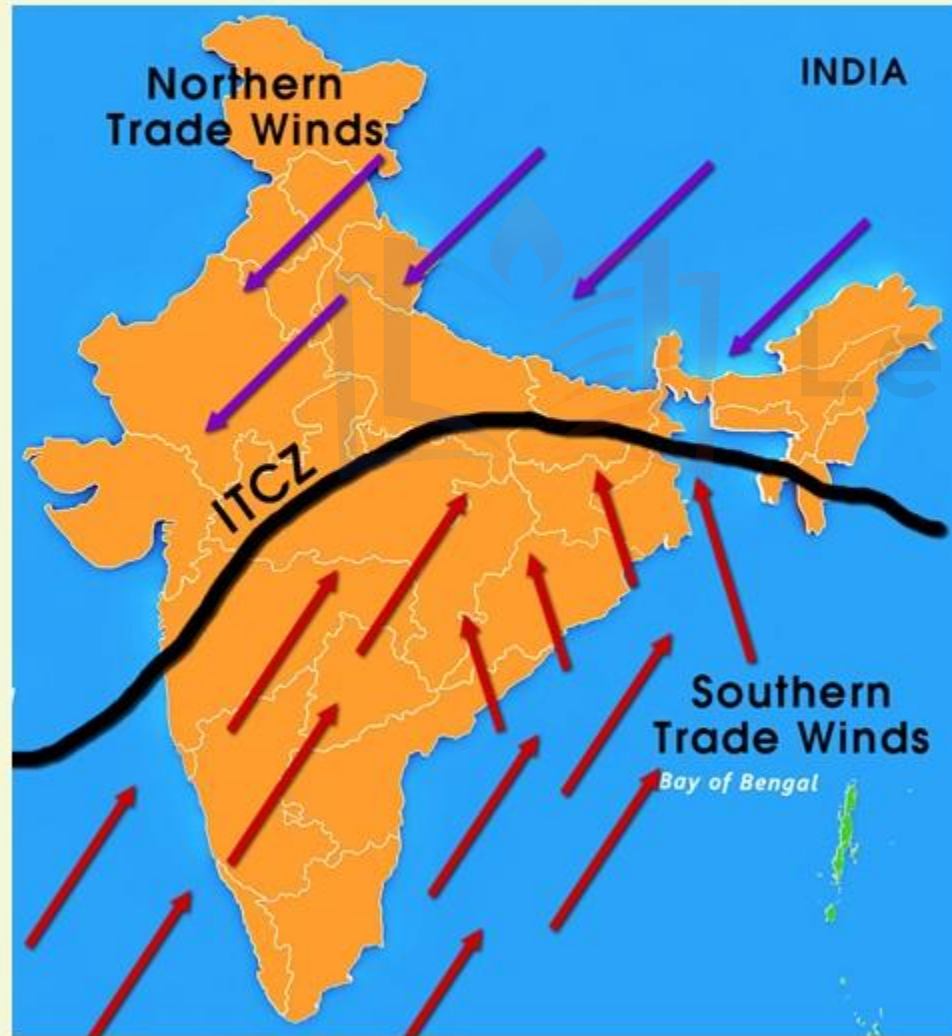
FACTORS RELATED TO AIR PRESSURE & WINDS

2

MECHANISM OF WEATHER IN THE SUMMER SEASON

FACTOR NO:1

SURFACE PRESSURE AND MONSOON WINDS



1

As the summer sets in and the sun shifts northwards, the wind circulation over the subcontinent undergoes a complete reversal at both, the lower as well as the upper levels.

2

By the middle of July, the low pressure belt nearer the surface termed as **Inter Tropical Convergence Zone (ITCZ)** shifts northwards, roughly parallel to the Himalayas between 20° N and 25° N.

3

By this time, the westerly jet stream withdraws from the Indian region.

FACTORS RELATED TO AIR PRESSURE & WINDS

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MECHANISM OF WEATHER IN THE SUMMER SEASON

FACTOR NO:1

SURFACE PRESSURE AND MONSOON WINDS

4 In fact, meteorologists have found an interrelationship between the northward shift of the equatorial trough (ITCZ) and the withdrawal of the westerly jet stream from over the North Indian Plain.

5 It is generally believed that there is a cause and effect relationship between the two.

6 The ITCZ being a zone of low pressure, attracts inflow of winds from different directions.

7 The maritime tropical airmass (mT) from the southern hemisphere, after crossing the equator, rushes to the low pressure area in the general southwesterly direction.

8 It is this moist air current which is popularly known as the southwest monsoon.

FACTORS RELATED TO AIR PRESSURE & WINDS

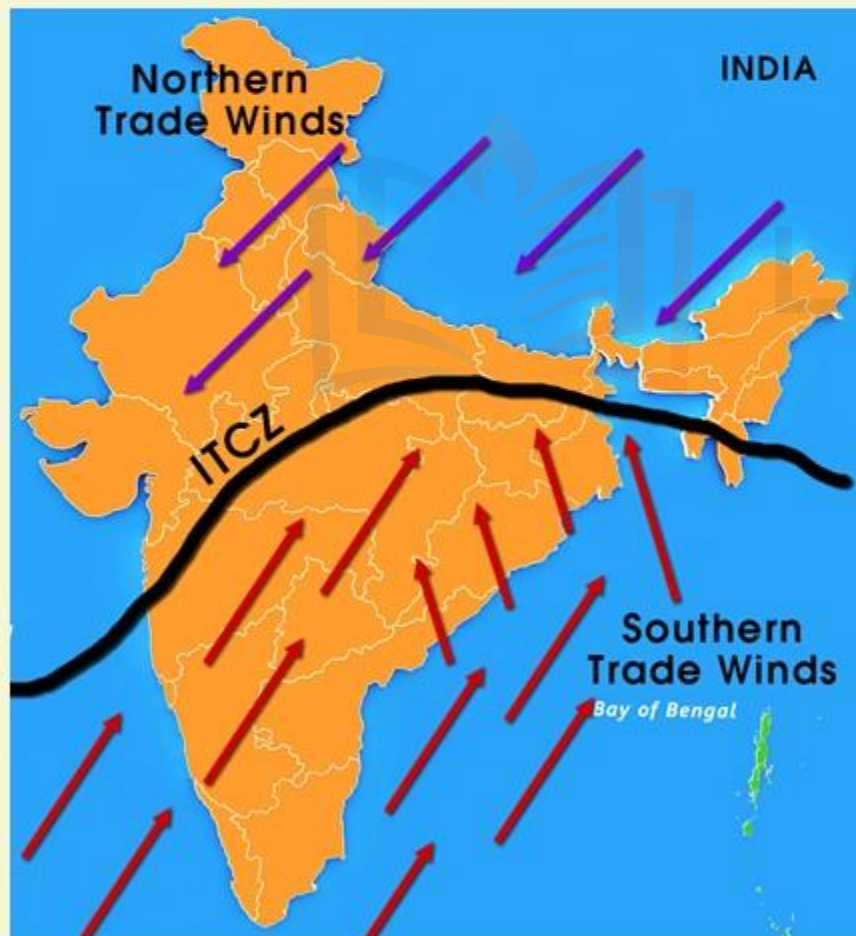
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MECHANISM OF WEATHER IN THE SUMMER SEASON

FACTOR NO:1

SURFACE PRESSURE AND MONSOON WINDS

INTER TROPICAL CONVERGENCE ZONE (ITCZ)



- The Inter Tropical Convergence Zone (ITCZ) is a **low pressure zone** located at the equator where trade winds converge, and so, it is a zone where air tends to ascend.
- In July, the ITCZ is located around 20°N - 25°N latitudes (over the Gangetic plain), sometimes called the monsoon trough.
- This **monsoon trough** encourages the development of thermal low over north and northwest India.

FACTORS RELATED TO AIR PRESSURE & WINDS

2

MECHANISM OF WEATHER IN THE SUMMER SEASON

FACTOR NO:1

SURFACE PRESSURE AND MONSOON WINDS

INTER TROPICAL CONVERGENCE ZONE (ITCZ)



- Due to the shift of ITCZ, the trade winds of the southern hemisphere cross the equator between 40° and 60° E longitudes and start blowing from southwest to northeast due to the Coriolis force.
- It becomes southwest monsoon.
- In winter, the ITCZ moves southward, and so the reversal of winds from northeast to south and southwest, takes place.
- They are called northeast monsoons.

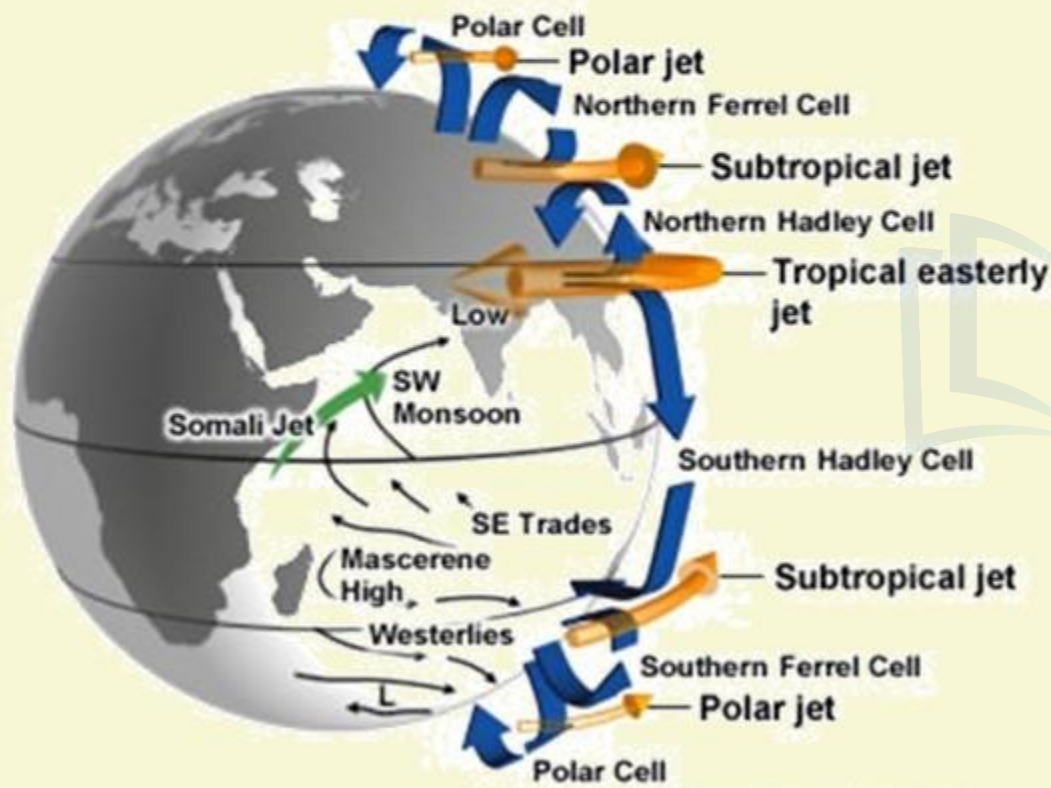
FACTORS RELATED TO AIR PRESSURE & WINDS

2

MECHANISM OF WEATHER IN THE SUMMER SEASON

FACTOR NO:2

JET STREAM AND UPPER AIR CIRCULATION



1 The pattern of pressure and winds as mentioned above is formed only at the level of the troposphere.

2 An easterly jet stream flows over the southern part of the Peninsula in June and has a maximum speed of 90 km per hour.

3 In August, it is confined to 15°N , latitude, and in September up to 22°N latitudes.

4 The easterlies normally do not extend to the north of 30°N latitude in the upper atmosphere.

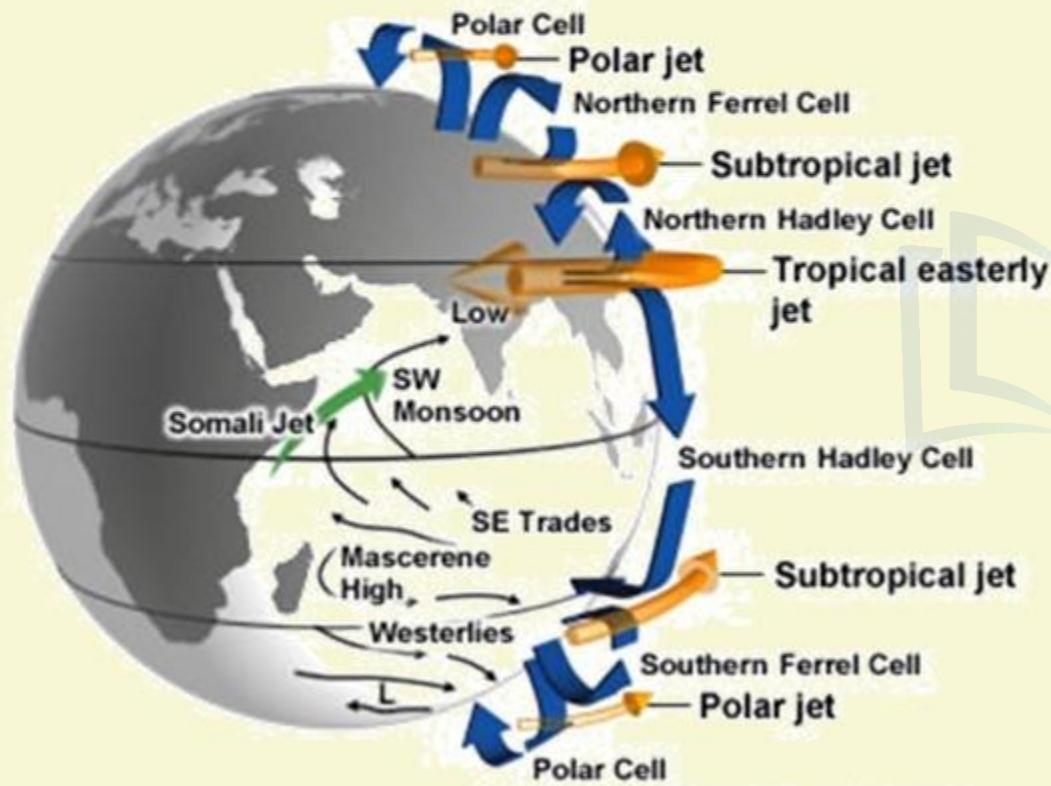
FACTORS RELATED TO AIR PRESSURE & WINDS

2

MECHANISM OF WEATHER IN THE SUMMER SEASON

FACTOR NO:2

EASTERLY JET STREAM & TROPICAL CYCLONES



- 1 The easterly jet stream steers the tropical depressions into India.
- 2 These depressions play a significant role in the distribution of monsoon rainfall over the Indian subcontinent.
- 3 The tracks of these depressions are the areas of highest rainfall in India.
- 4 The frequency at which these depressions visit India, their direction and intensity, all go a long way in determining the rainfall pattern during the southwest monsoon period.

FACTORS DETERMINING THE CLIMATE OF INDIA

India's climate is controlled by a number of factors which can be broadly divided into two groups:

1. Factors related to **location and relief**
2. Factors related to **air pressure and winds**.

POINT NO:2

FACTORS RELATED TO AIR PRESSURE & WINDS

These factors related to air pressure and winds can be further sub-divided into the following:

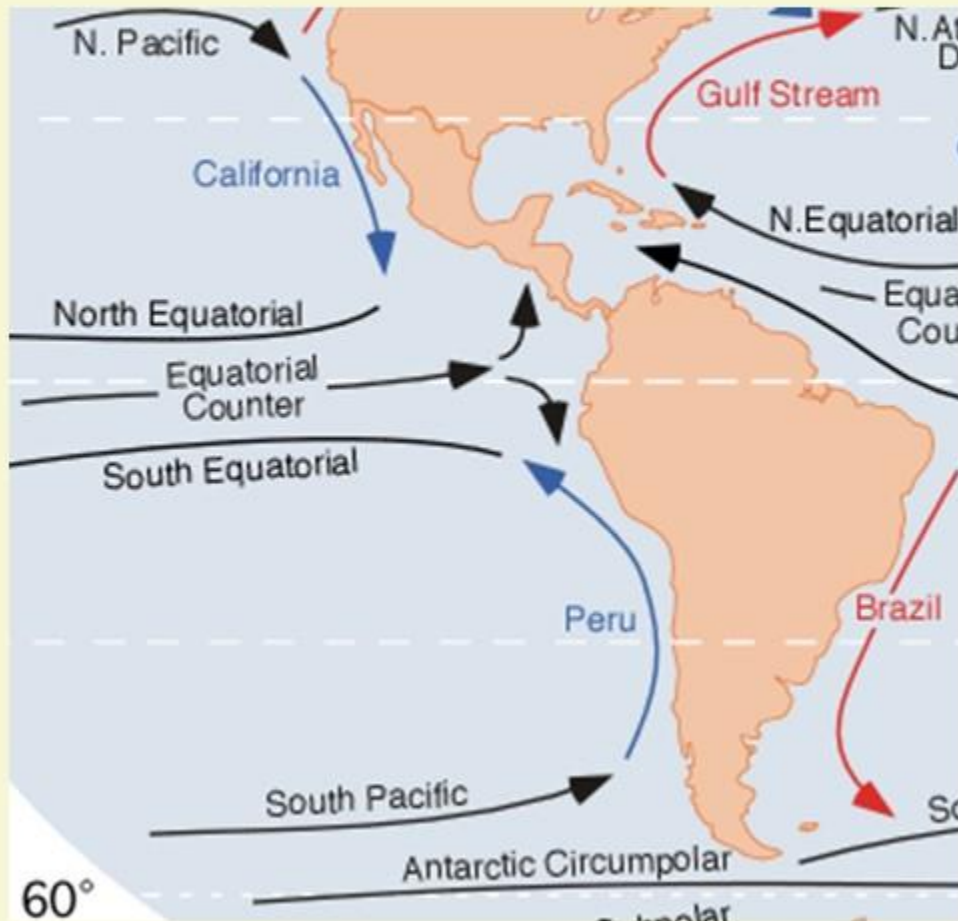
1. Monsoon wind movement
2. Upper Air Circulation- Easterly and Westerly Jetstream
3. Tropical cyclones and Western Disturbances
4. El Nino effect
5. La Nina effect
6. Southern Oscillation

**EFFECT DURING
SUMMER AND
WINTER SEASON**

FACTORS RELATED TO AIR PRESSURE & WINDS

FACTOR NO:4

EI-NINO AND THE INDIAN MONSOON

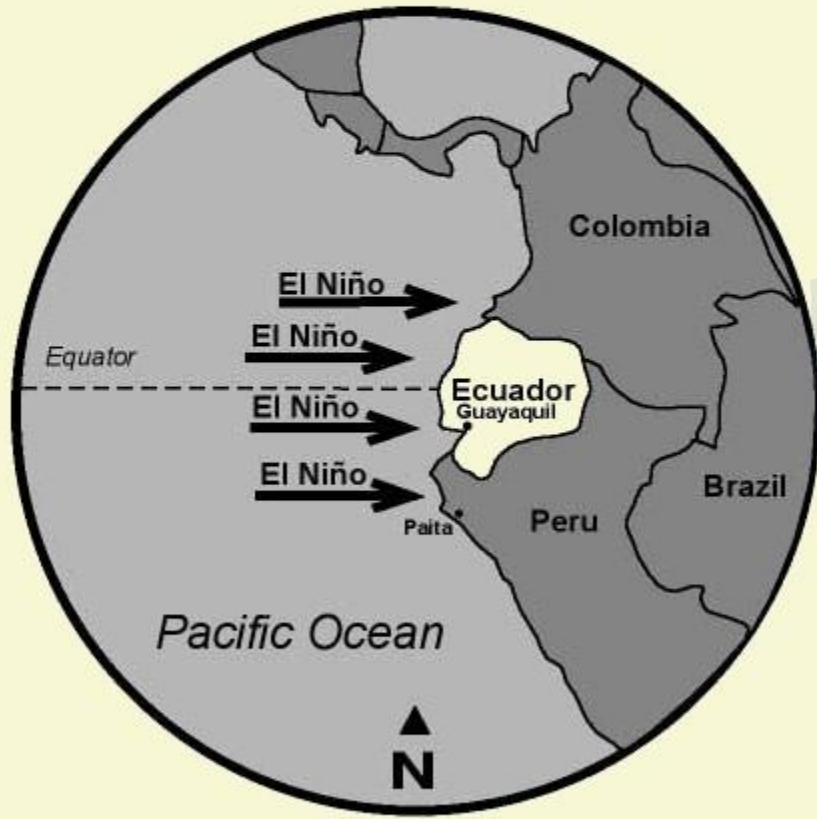


- El-Nino is a complex weather system that brings drought, floods and other weather extremes to different parts of the world.
- The system involves oceanic and atmospheric phenomena with the appearance of warm currents off the coast of Peru in the Eastern Pacific and affects weather in many places including India.
- El-Nino is merely an extension of the warm equatorial current which gets replaced temporarily by cold Peruvian current or Humbolt current (locate these currents in your atlas).

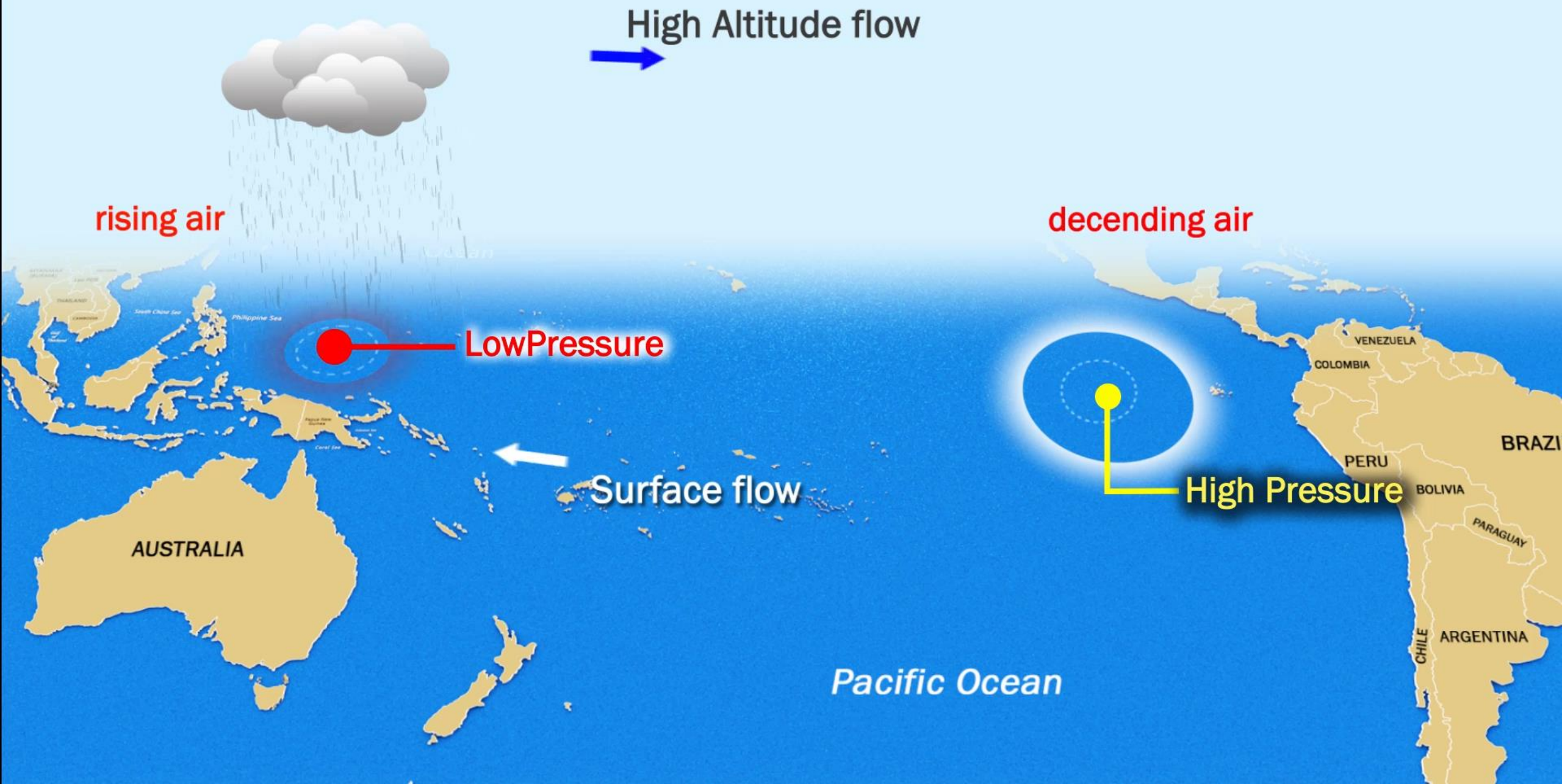
FACTORS RELATED TO AIR PRESSURE & WINDS

FACTOR NO:4

EI-NINO AND THE INDIAN MONSOON



- The word **El-Nino** means 'Child Christ' because this current appears around Christmas in December. December is a summer month in Peru (Southern Hemisphere).
- El-Nino is used in India for forecasting long range monsoon rainfall.
- In 1990-91, there was a wild El-Nino event and the onset of southwest monsoon was delayed over most parts of the country ranging from five to twelve days.



FACTORS RELATED TO AIR PRESSURE & WINDS

FACTOR NO:4

EI-NINO AND THE INDIAN MONSOON

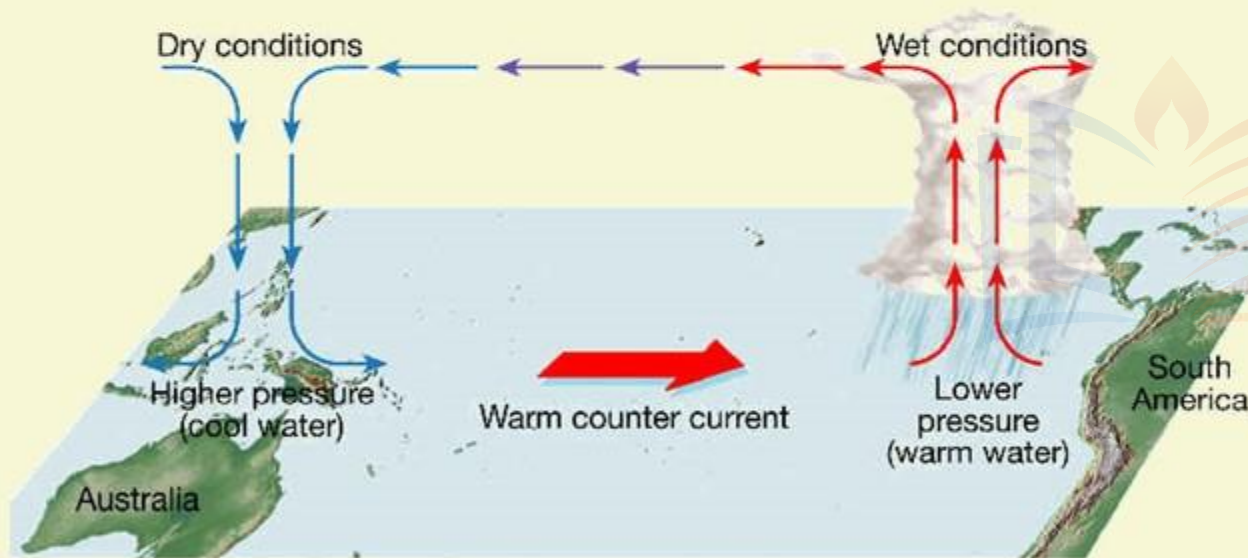


- 1 El-Nino is a narrow warm current which occasionally appears off the coast of Peru in December.
- 2 It is a temporary replacement of the cold Peru current, which normally flows along the coast.
- 3 This current is responsible for wide spread floods and droughts in the tropical regions of the world.

FACTORS RELATED TO AIR PRESSURE & WINDS

FACTOR NO:4

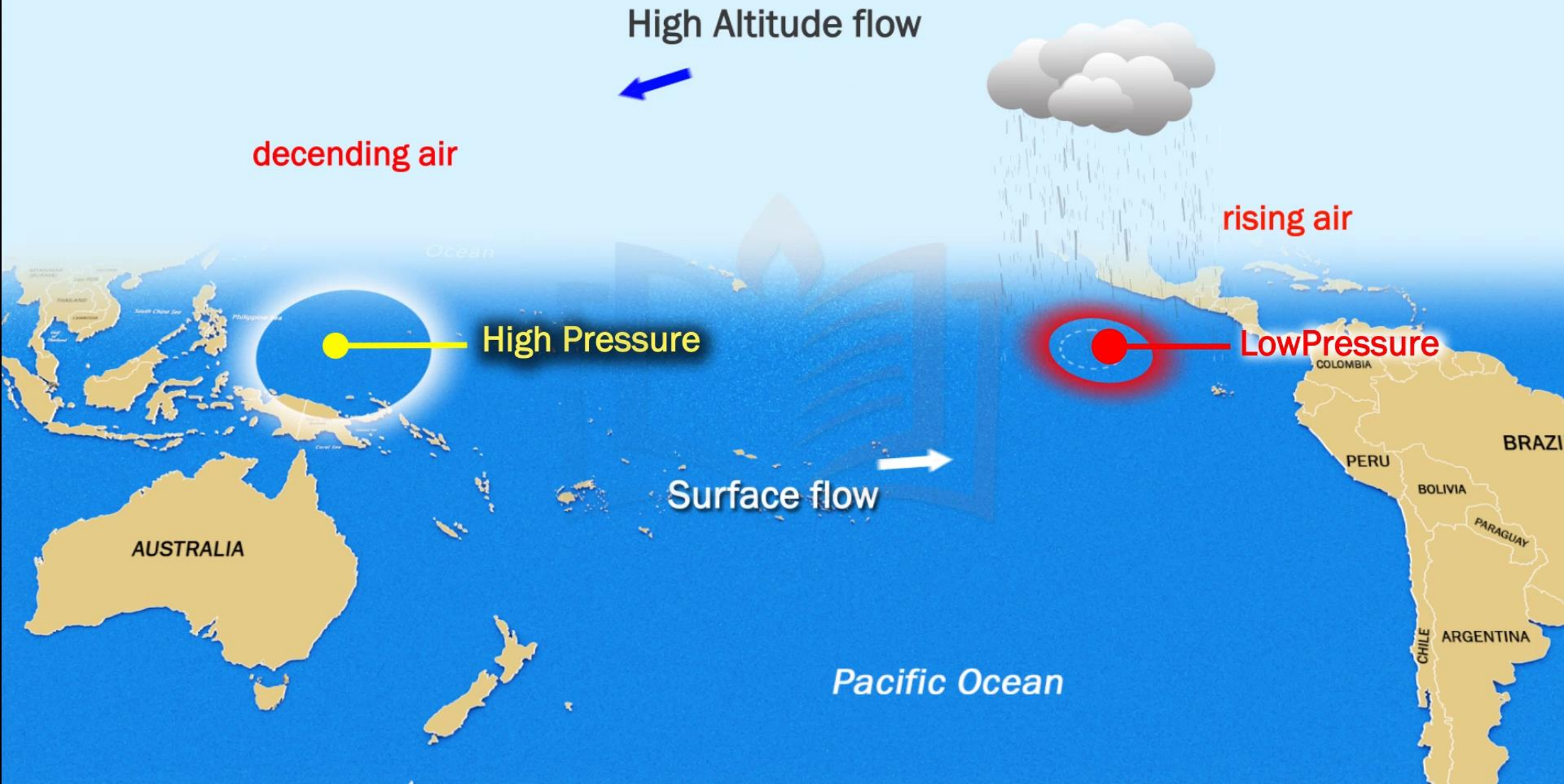
EI-NINO AND THE INDIAN MONSOON



4 Sometimes it becomes more intense and increases the surface water temperatures of the sea by 10^0 C.

5 This **warming of tropical Pacific waters** affects the global pattern of pressure and wind systems including the monsoon winds in the Indian Ocean.

6 Meteorologists believe that the severe drought of 1987 in India was caused by El-Nino.



FACTORS RELATED TO AIR PRESSURE & WINDS

FACTOR NO:4

LA-NINA AND THE INDIAN MONSOON

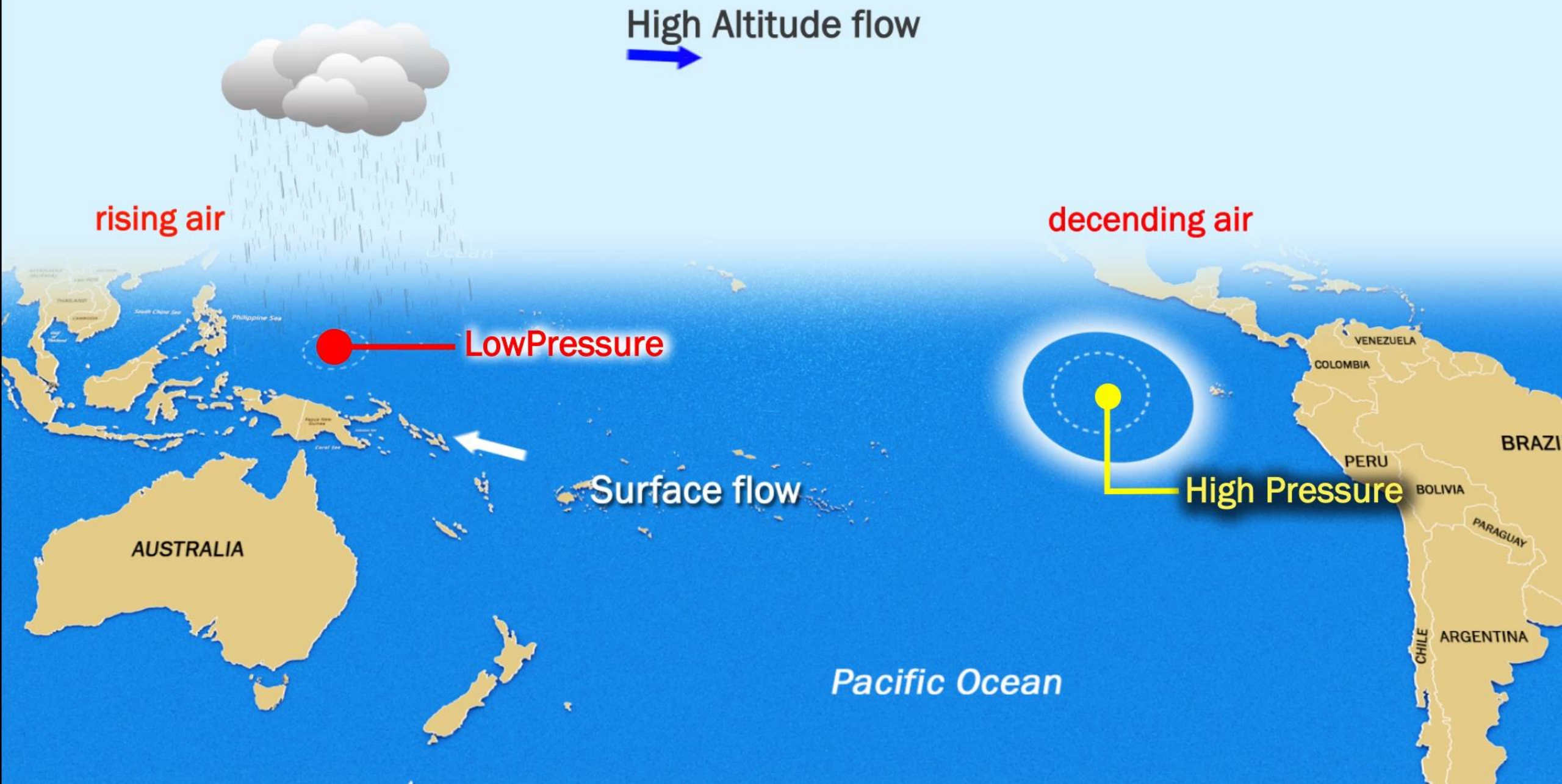
1 After an El-Nino, weather conditions return to normal.

2 However, sometimes trade winds become so strong that they cause abnormal accumulation of cold water in the central and eastern Pacific region.

3 This event is called La Nina, which in effect is the complete opposition of EL Nino.

4 A La Nina also marks an active hurricane season.

5 But, in India, the presence of La Nina portends exceptionally good news. It is the harbinger of heavy monsoon showers in India.

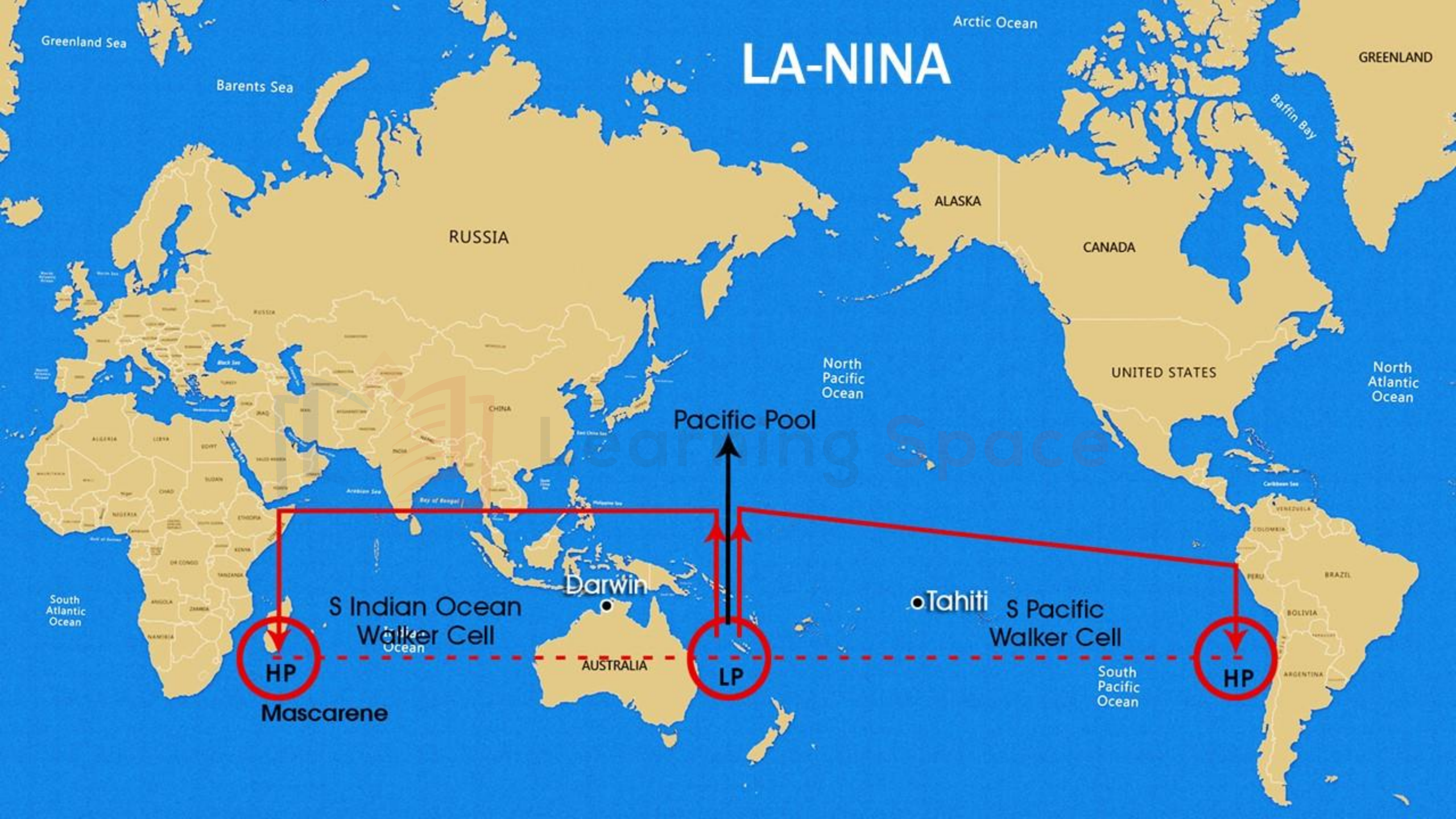


FACTORS RELATED TO AIR PRESSURE & WINDS

FACTOR NO:5

LA-NINA AND THE INDIAN MONSOON

- 1 There is a strange linkage of meteorological changes often observed between the Indian and Pacific Oceans.
- 2 It has been noticed that whenever the surface level pressure is high over the Indian Ocean, there is low pressure over the Pacific Ocean and Vice-versa.
- 3 This interrelation of high and low pressure over the Pacific and the Indian Ocean is called Southern Oscillation.
- 4 When the winter pressure is high over the Pacific Ocean and low over the Indian Ocean, the south-west monsoons in India tend to be stronger.
- 5 In the reverse case, the monsoons are most likely to be weaker.



FACTORS RELATED TO AIR PRESSURE & WINDS

FACTOR NO:5

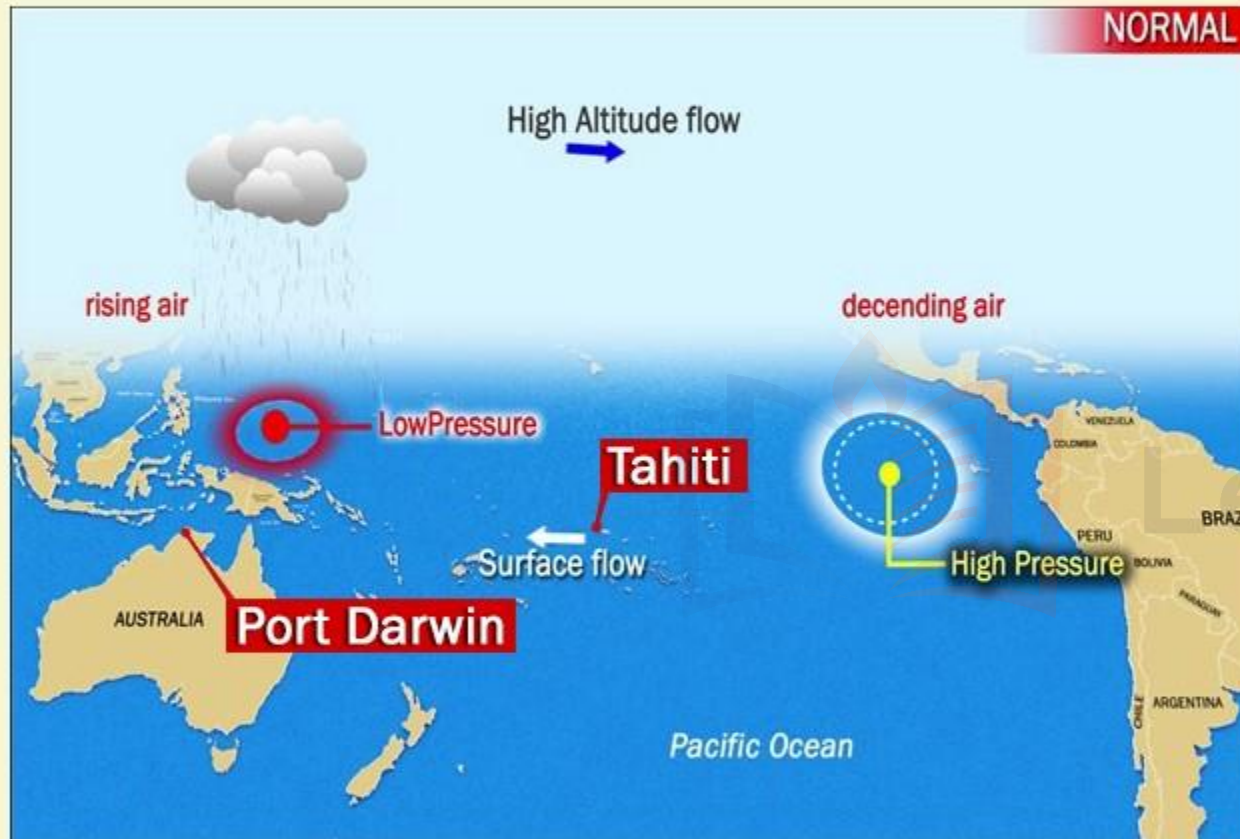
EL NINO & SOUTHERN OSCILLATION [ENSO]

- 1** The formation of an El Niño is linked with Pacific Ocean circulation pattern known as the southern oscillation [circulation of atmospheric pressure]
- 2** Southern Oscillation is a coherent inter-annual fluctuation of atmospheric pressure over the tropical Indo-Pacific region.
- 3** El Nino and Southern Oscillation coincide most of the times hence their combination is called ENSO– El Nino Southern Oscillation.
- 4** The pattern of low and high pressures gives rise to vertical circulation with surface flow and atmospheric flow leading to Southern Oscillations and due to its close association with El-Nino development, it is combinedly called as ENSO event.
- 5** Thus, ENSO can be said to be a best example of atmospheric-oceanic telecommunication.

FACTORS RELATED TO AIR PRESSURE & WINDS

FACTOR NO:5

EL NINO & SOUTHERN OSCILLATION [ENSO]



- 1 Southern Oscillation Index (SOI) is used to measure the **intensity of the Southern Oscillation**.
- 2 This is the difference in pressure between **Tahiti in French Polynesia** representing the Central Pacific Ocean and **Port Darwin in northern Australia** representing the Western Pacific Ocean.
- 3 The positive and negative values of the SOI i.e. Tahiti minus the Port Darwin pressure are pointers towards good or bad rainfall in India.

FACTORS RELATED TO AIR PRESSURE & WINDS

FACTOR NO:5

EL NINO & SOUTHERN OSCILLATION [ENSO]

POSITIVE SOI

Tahiti pressure is greater than Port Darwin

Drought conditions in East Pacific and good rainfall in West Pacific (Northern Australia and Indonesia)

Good for Indian Monsoons

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